

**2017**

**MBA**

**3rd Semester Examination**

**SECURITY ANALYSIS AND PORTFOLIO MANAGEMENT**

**(Specialisation : Financial Management)**

**PAPER—F-307**

*Full Marks : 100*

*Time : 3 Hours*

*The figures in the right-hand margin indicate full marks.*

*Candidates are required to give their answers in their own words as far as practicable.*

*Illustrate the answers wherever necessary.*

1. Answer any *eight* questions of the following : 8×5
- (a) What is speculation ? How does it differ from gambling ? 2½+2½
- (b) Discuss the Random Walk theory.

*(Turn Over)*

- (c) Explain the term 'yield to maturity' (YTM) in the case of a bond ? Calculate the YTM of a 12% par bond (face value is Rs. 1000) which will mature after five years. 2+3
- (d) What do you understand by stock market indices ? Write a short note on Nifty.
- (e) The following information is given regarding the return from a security A and a market index :

| Month     | Monthly return |                |
|-----------|----------------|----------------|
|           | From A(%)      | From Market(%) |
| April     | 3              | 2              |
| May       | 5              | 1.7            |
| June      | 1              | 2              |
| July      | 4              | 3              |
| August    | (2)            | (3)            |
| September | 2.50           | 1.80           |
| October   | 2.1            | 2.5            |
| November  | 1.5            | 1.7            |

You are required to compute the beta of the security and comment on the result.

- (f) In the context of fundamental analysis, discuss the company level of analysis.

- (g) Explain the importance of the CAPM. Using the model, calculate the expected return from security E having a beta of 2.15 when it is given that the expected return from two securities C and D, having beta of 1.40 and 1.80 are 15% and 18% respectively.
- (h) Discuss the role of SEBI in protecting the interest of investors in the capital market.
- (i) What are the differences between open-ended and close-ended funds ?
- (j) Discuss the objectives of portfolio management.
- (k) Explain the terms 'value investing' and 'growth investing'.
- (l) Explain the sources of risk in relation to investment in a security.

2. Answer any *four* of the following :

4×10

(a) Given the following—

|                       | Security ABC | Security XYZ |
|-----------------------|--------------|--------------|
| Expected Return (%)   | 40           | 25           |
| Expected Variance (%) | 25           | 16           |

State whether a risk averse will have any diversification benefit informing portfolio of above two securities in accordance with Marketing diversification principle if the covariance between two securities' return happen to be either 16 or  $(-)$  20. If you recommend formation of portfolio in any of the two cases, determine the minimum variance portfolio combination and the risk-return profile for such combination.

10

(b) (i) What are the advantages of investing in a mutual fund ?

(ii) The earnings and dividends of Star Ltd. are currently growing at 21%. This may continue for the next 5 years and there after the growth rate will reduce to 10% indefinitely. The dividend paid in the current year is 32%. The share of the company trades at Rs. 57 per share (face Value Rs. 10 per share). Would you buy the stock if your required rate of return is 20%.

4+6

(c) (i) The following information is given for three Bankex funds —

| Mutual fund | Average yearly return(%) | Beta value |
|-------------|--------------------------|------------|
| Sona        | 15.0                     | 1.40       |
| Mona        | 19.8                     | 1.26       |
| Dona        | 23.9                     | 1.78       |

It is also mentioned that the risk-free rate of return is 5% and return on Bankex in the last year was 14.9%. You are required to evaluate and rank the funds using Jensen's performance measure.

- (ii) The equity shares of Chamatkar Ltd. (face value is Rs. 10) are listed on the NSE. The company declared a dividend of Rs. 6.20 in 2014 and dividends have been growing at the rate of 12% since then which shall continue till infinity. Should you buy the share if it is currently trading at Rs. 255 ? Assume that the cost of capital is 15%. Show necessary calculations. 6+4
- (d) (i) Explain the concept of charts in the context of technical analysis.
- (ii) The following is the closing price of the share of ABC Ltd. during the last 16 days :

| Day | Rs. | Day | Rs. |
|-----|-----|-----|-----|
| 1   | 98  | 9   | 103 |
| 2   | 96  | 10  | 106 |
| 3   | 94  | 11  | 101 |
| 4   | 103 | 12  | 106 |
| 5   | 100 | 13  | 107 |
| 6   | 99  | 14  | 110 |
| 7   | 95  | 15  | 107 |
| 8   | 101 | 16  | 109 |

You are required to compute the relative 7-day relative strength index. Why such an index is called an oscillator ?

4+(4+2)

- (e) (i) Write down the formulae for portfolio risk in the case of a three-security portfolio.
- (ii) The following is the expected data regarding the states of the economy and the return from two securities G and H.

| State of the economy | Pr obability | Re turn of G (%) | Re turn of H (%) |
|----------------------|--------------|------------------|------------------|
| Good                 | 40%          | 14               | 15               |
| Moderate             | 30%          | 12               | 11               |
| Bad                  | 30%          | 8                | 7                |

You are required to compute the following :

- Expected return of G and H
- Correlation coefficient between the two securities.

2+(2+6)

- (f) (i) What are the basic characteristics that differentiate securities ?
- (ii) Explain the term 'diversification' in the context of portfolio management. Discuss the role of correlation coefficient between securities in reducing portfolio risk through diversification.

3+(3+4)

**[ Internal Assessment : 20 marks ]**